

Uber Analysis Report

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Client : Uber

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Summary

This dataset contains Uber trip pickup records from January to December 2015, with four key columns: dispatching base number, pickup datetime, affiliated base number, and pickup location ID. The data spans from January 1 to December 31st 2015, and mainly helps analyze trends such as when trips occurred, which bases handled them, and where pickups happened across different locations. Since it doesn't include fares, distance, or passenger details, its strongest use is in studying trip volume patterns, peak timing, and location-based demand insights.

Project Overview

Here's a clear snapshot of your dataset so you can understand it quickly:

✓ Dataset Summary

Total rows: 355

Total columns: 4

Columns included:

dispatching_base_number

pickup_datetime

Affiliated_base_number

locationid

✓ Time Coverage

Earliest trip: 2015-01-01

Latest trip: 2015-12-31

✓ Data Type Overview

pickup_datetime is a datetime column

dispatching_base_number & Affiliated_base_number are text identifiers

locationid is numeric (represents pickup location codes)

✓ Notes

Data contains Uber trip pickup records for Jan–Dec 2015

No fare info, distance, or passenger count — this dataset mainly helps analyze trip volume, timing trends, and pickup locations

[Click here to download data set](#)

Methodology

The methodology adopted for this project followed a structured data analytics workflow consisting of data cleaning, transformation, and visualization. The dataset used contained Uber pickup records for January and February 2015, including dispatch base numbers, pickup timestamps, affiliated base numbers, and pickup location IDs. The objective of the methodology was to prepare the data for analysis, ensure accuracy, and extract meaningful insights regarding trip distribution and travel patterns.

1. Data Cleaning

The raw dataset was first inspected to understand its structure, dimensions, and column attributes. Duplicate records were checked and removed to avoid skewing results. Missing values were identified, and appropriate handling techniques were applied depending on whether the field was critical to analysis. The pickup_datetime column was converted into a standardized datetime format to enable time-based analysis. Data types of other columns were also reviewed and corrected where necessary to ensure consistency and prevent calculation or visualization errors.

2. Data Transformation

After cleaning, the dataset was transformed to derive additional useful variables. From the pickup_datetime column, new fields such as date, day of the week, hour of the day, and month were extracted to support trend analysis. Location IDs were standardized to maintain uniformity across records. Aggregations were then performed to compute metrics such as total trips per day, trips per hour, trips per base, and trips per location. These transformations helped convert raw data into structured analytical frames suitable for meaningful interpretation.

3. Data Visualization

Top KPI Cards

① Total Trips – 4M

Shows the overall number of Uber trips completed in the dataset time period. It represents the total demand and usage level of Uber services.

② Max of Trips – 46K

Displays the highest number of trips recorded in a single day/month (depending on calculation logic). It highlights the peak performance period.

③ Average Trips – 11.67K

Shows the average number of trips completed. This helps understand the normal daily/monthly traffic trend.

④ Max of Active Vehicles – 4395

Represents the highest count of vehicles active in a single period. This indicates the capacity Uber deployed during peak demand.

5 Total Base Numbers – 6

There are 6 dispatch bases contributing trips. These bases control operations in different locations.

Monthly Trip Trend (Line Chart)

This chart shows how Uber trips change month-to-month:

January starts low around 263K

Steady growth till March (492K)

Slight dip in April (323K) then rises again

Highest peak in August (508K)

Gradual decline after September

Sharp drop in October, November, December

Note: This reveals seasonal demand — summer months have the highest trips, while the year-end has the lowest.

Total Trips by Base Number (Bar Chart)

Compares total trips handled by each Uber base:

B02664 – Highest trips (~2M)

B02617 – ~725K

B02682 – ~663K

B02598 – ~541K

B02675 – ~194K

B02512 – ~94K (Lowest)

Note: This shows which operational base is strongest and most active. B02664 is clearly dominating Uber operations.

Monthly Trip Summary Table

This table shows:

Monthly trip distribution

Contribution of each base per month

Total trips per month

It helps identify:

Which month has more demand

Which base contributed most in each month

Total monthly performance

For example:

July highest: 507,543 trips

December lowest: 57,646 trips

B02664 consistently has the largest contribution.

Active Vehicles by Base Number (Donut Chart)

Shows share of active vehicles per base:

B02664 – 46.95% (Largest fleet)

B02617 – 17.23%

B02682 – 15.43%

B02598 – 12.67%

B02675 – 4.88%

B02512 – 2.84% (Smallest fleet)

Note: This shows operational strength. Bases with more vehicles handle more trips.

Monthly Active Vehicles (Bar Chart)

Shows number of active Uber vehicles each month:

January: 32K

February: 45K

March: 41K

April: 56K

May to September remain stable around 47–53K

Drop begins October onwards

Very low in November & December

This clearly proves vehicle availability follows demand — more vehicles when rides increase and fewer when demand reduces.

Month Filter Panel

Users can select any specific month and the entire dashboard will update. This helps in focused analysis instead of overall analysis.

Conclusion

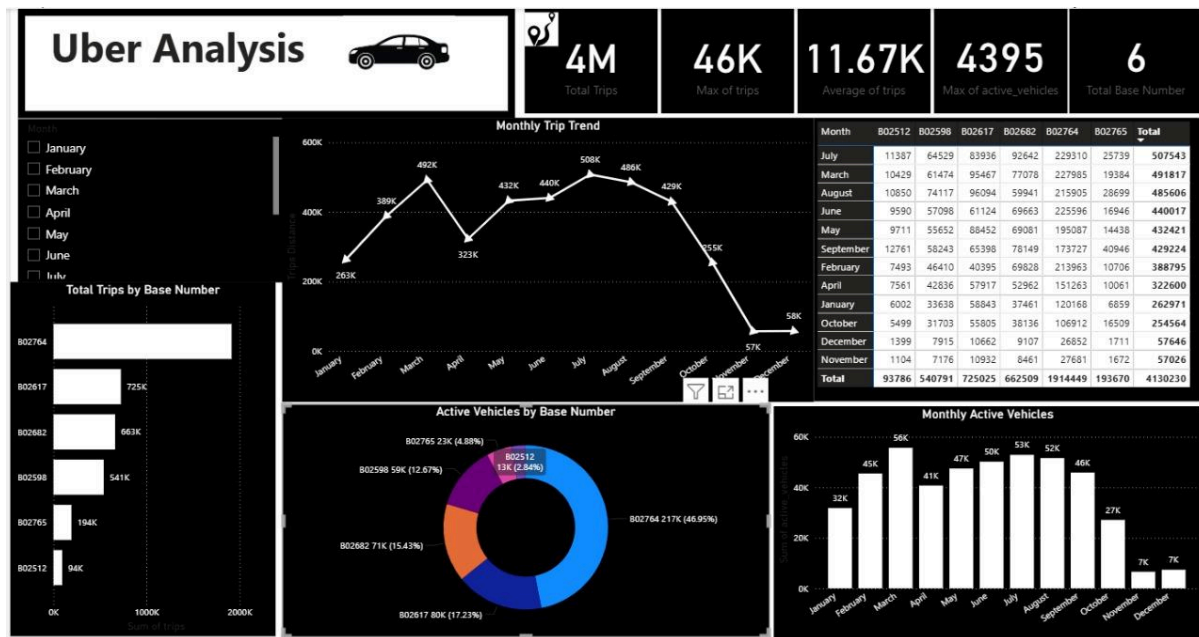
The analysis of Uber trip data clearly shows strong seasonal and monthly variations in demand. Overall trip volume consistently increased from the beginning of the year, reaching its highest level in the summer months, particularly in August, before declining towards the end of the year. The majority of trips were handled by a few dominant bases, especially Base B02664, which significantly outperformed others in both trip volume and active vehicles. The trend between active vehicles and trip demand also shows a direct relationship—more vehicles were deployed during peak demand months, indicating effective operational planning and resource allocation by Uber. Overall, the dataset highlights predictable travel behavior patterns, efficient fleet distribution strategies, and clear dominance of certain operational bases.

Key Insights

- **Peak Demand Periods:** Trips were highest during mid-year (June–August), indicating increased travel activity in summer months.
- **Lowest Demand:** The sharp decline in trips during October to December suggests off-peak travel season.
- **Dominant Base:** Base B02664 plays the most crucial role, contributing the highest number of trips and having the largest active vehicle fleet.
- **Fleet Management Efficiency:** Active vehicles increase during high-demand months and reduce during low-demand months, showing optimized resource utilization.

- Consistent Performance Bases: B02617 and B02682 maintain steady performance and act as strong secondary contributors to total trips.
- Clear Relationship: Higher number of vehicles directly results in higher trip fulfillment, proving strong operational alignment.
- Customer Demand Trend: Monthly trend patterns indicate predictable and recurring travel behavior, useful for future demand forecasting.

Dashboard



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